21/02/24 M/S

USN | | | | | | | | | | [22TCSE116]

FOURTH SEMESTER B.Tech EXAMINATION, FEBRUARY-2024
ESSENTIALS OF DEEP LEARNING

Time: 3 Hours **Maximum Marks: 100**

Instructions: Assume missing data

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define Feed forward Neural Network.	L-1	CO1	**02**
	b.	List out some advantages of multilayer perceptron.	L-1	C02	**02**
	c.	Define cross entropy loss.	L-1	C03	**02**
	d.	Define deep learning with its uses.	L-1	C04	**02**
	e.	Define Black-Box attack	L-1	C05	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Describe briefly about characteristics of deep learning and mention how it differs from machine learning.	L2	C01	**05**
		OR			
	b.	Explain briefly about types of deep learning algorithms and its advantages.	L2	C01	**05**
3.	**a.**	Explain neatly about perceptron with its structure diagram.	L2	C02	**05**
		OR			
	b.	Discuss about activation function and explain its working method in neural networks.	L2	C02	**05**
4.	**a.**	Discuss briefly about key advantages of using the pooling layer in neural network problems.	L2	C03	**05**
		OR			
	b.	Explain briefly about mobile net architecture in neural networks?	L2	C03	**05**
5.	**a.**	What is Recurrent neural network, and explain with an application?	L2	C04	**05**
		OR			
	b.	Explain briefly about different applications in Generative model.	L2	C04	**05**
6.	**a.**	Distinguish between Generative and discriminative models.	L4	C05	**05**
		OR			
	b.	What are the types of adversarial attacks? Explain in brief.	L2	C05	**05**
7.	**a.**	What is convolutional neural network and explain with any two applications?	L2	C04	**05**
		OR			

	b.	Compare and contrast between AlexNet and LeNet.	L4	C04	**05**
		PART-C	**06 X 10=60**		
8.	**a.**	Discuss briefly about the basic building block of neural networks with diagram.	L2	C01	**10**
		OR			
	b.	Explain the working method of backpropogation concept and mention the advantage of using backpropogation in neural networks.	L2	C01	**10**
9.	**a.**	What is multilayer perceptron? Mention its characteristics of performing output prediction in neural networks.	L2	C02	**10**
		OR			
	b.	Describe briefly about entropy loss function in neural networks.	L2	C02	**10**
10.	**a.**	Explain briefly about characteristics of convolutional neural network in image analysis.	L2	C03	**10**
		OR			
	b.	Elaborate a sketch on different types of CNN architecture and their advantages.	L2	C03	**10**
11.	**a.**	Mention the key advantages of using the Seq2Seq models in neural network problems	L2	C04	**10**
		OR			
	b.	What is Recurrent neural network and explain with an application?	L2	C04	**10**
12.	**a.**	Describe briefly about types of RNN based upon number of inputs and outputs in the network.	L2	C05	**10**
		OR			
	b.	Write short notes on **i.** Seq2Seq models **ii.** Attention mechanism.	L1	C05	**10**
13.	**a.**	Explain briefly about the working concepts of artificial neural network with diagram.	L2	C04	**10**
		OR			
	b.	Describe briefly about discriminative model explain how it differs from generative model.	L2	C05	**10**